

Mechanical Department AI Training — Question & Answer Dataset

Prepared as a structured knowledge reference for a Mechanical Department educational AI. These Q&A; pairs are suitable as source material for RAG/knowledge grounding or for preparing a fine-tuning dataset.

1. Centre Lathe & Lathe Operations

Q1. What is a centre lathe?

Answer: A centre lathe is a machine tool in which a workpiece rotates while a cutting tool removes material to produce cylindrical and other surfaces.

Q2. What is facing in a lathe?

Answer: Facing is a lathe operation used to produce a flat surface on the end of a workpiece.

Q3. What is turning?

Answer: Turning is the operation of reducing the diameter of a rotating workpiece by feeding a cutting tool along its axis.

Q4. What is taper turning?

Answer: Taper turning produces a conical surface whose diameter changes uniformly along the length of the workpiece.

Q5. What is boring?

Answer: Boring enlarges or improves the accuracy of an existing hole using a single-point cutting tool.

Q6. What is drilling on a lathe?

Answer: Drilling produces a new cylindrical hole using a drill held in the tailstock or another suitable holder.

Q7. What is knurling?

Answer: Knurling is a forming operation that produces a regular patterned surface by pressing hardened rollers against a rotating workpiece.

Q8. What is threading?

Answer: Threading produces a helical thread on the external or internal surface of a workpiece.

Q9. What is chamfering?

Answer: Chamfering removes a sharp edge and produces a small angled surface at the edge of a workpiece.

Q10. What is parting-off?

Answer: Parting-off is a lathe operation used to cut a finished part from the parent workpiece.

Q11. What is contour turning?

Answer: Contour turning produces a curved or irregular profile by controlling the tool path along the workpiece.

Q12. What is the function of a chuck?

Answer: A chuck holds and rotates the workpiece securely in the lathe spindle.

Q13. What is the function of the tailstock?

Answer: The tailstock supports long workpieces with a centre and can hold tools such as drills.

Q14. What is cutting speed?

Answer: Cutting speed is the relative speed between the cutting edge and the surface of the workpiece at the cutting zone.

Q15. What is feed?

Answer: Feed is the distance the cutting tool advances relative to the workpiece for a specified amount of cutting.

Q16. What is depth of cut?

Answer: Depth of cut is the thickness of material removed in one pass, measured approximately perpendicular to the machined surface.

Q17. Why is coolant used in machining?

Answer: Coolant helps remove heat, reduce friction, improve tool life, and flush chips away from the cutting zone.

Q18. What is a cutting tool?

Answer: A cutting tool is a hardened tool with a cutting edge used to remove material from a workpiece.

Q19. What is a centre in a lathe?

Answer: A centre supports a workpiece at its axis, especially during machining of long components.

Q20. What is the purpose of the carriage?

Answer: The carriage supports and moves the cutting tool along the lathe bed.

2. Arc Welding

Q21. What is arc welding?

Answer: Arc welding is a fusion welding process in which heat from an electric arc melts the joint area and usually a filler electrode.

Q22. What is an electric arc?

Answer: An electric arc is a sustained electrical discharge between the electrode and workpiece that produces intense heat.

Q23. What is an electrode?

Answer: An electrode conducts welding current and may also supply filler metal to the joint.

Q24. What is SMAW?

Answer: SMAW, or Shielded Metal Arc Welding, uses a flux-coated consumable electrode to create the weld.

Q25. Why is flux used in welding electrodes?

Answer: Flux helps protect the molten weld pool from atmospheric contamination and forms slag over the weld.

Q26. What is a weld bead?

Answer: A weld bead is the deposited strip of solidified weld metal produced by a welding pass.

Q27. What is a butt joint?

Answer: A butt joint joins two pieces placed approximately in the same plane with their edges meeting.

Q28. What is a lap joint?

Answer: A lap joint joins two overlapping pieces by welding along the overlap.

Q29. What is a fillet weld?

Answer: A fillet weld joins two surfaces approximately at an angle, commonly in a T, lap, or corner joint.

Q30. What is welding current?

Answer: Welding current is the electrical current flowing through the welding circuit and strongly affects heat input and penetration.

Q31. Why is a welding helmet required?

Answer: A welding helmet protects the eyes and face from arc radiation, sparks, and hot particles.

Q32. What is weld penetration?

Answer: Weld penetration is the depth to which the weld extends into the base material or joint.

Q33. What is porosity in welding?

Answer: Porosity is the presence of gas cavities or pores trapped in the solidified weld metal.

Q34. What is slag?

Answer: Slag is a non-metallic layer formed from flux during welding and normally removed after the weld cools.

Q35. Why must the workpiece be properly grounded?

Answer: Proper grounding provides a safe electrical return path and helps maintain stable welding operation.

3. Shaping Machine

Q36. What is a shaping machine?

Answer: A shaping machine removes material using a reciprocating single-point cutting tool to produce mainly flat surfaces.

Q37. What is the ram?

Answer: The ram is the reciprocating member that carries the tool head and moves the cutting tool over the workpiece.

Q38. What is the tool head?

Answer: The tool head holds the cutting tool and can be adjusted for machining operations.

Q39. What is the return stroke?

Answer: The return stroke is the stroke during which the ram moves back and the tool normally does not perform the main cutting action.

Q40. What is quick return motion?

Answer: Quick return motion makes the return stroke faster than the cutting stroke to reduce idle time.

Q41. What surfaces can a shaper produce?

Answer: A shaper can produce horizontal, vertical, and inclined flat surfaces, slots, and similar profiles.

Q42. What is a clapper box?

Answer: The clapper box allows the tool to lift slightly during the return stroke, reducing rubbing between tool and workpiece.

Q43. What is feed in a shaper?

Answer: Feed is the controlled movement of the workpiece or table relative to the cutting tool between successive strokes.

Q44. Why is the shaper suitable for small jobs?

Answer: Its simple setup and reciprocating cutting action make it useful for producing flat surfaces and slots on small or medium workpieces.

Q45. What is the difference between a shaper and a planer?

Answer: A shaper generally moves the cutting tool while the workpiece is stationary or moves for feed; a planer generally reciprocates the workpiece while the tool performs the cutting.

4. Planing Machine

Q46. What is a planer?

Answer: A planer is a machine tool mainly used to machine large and heavy workpieces by reciprocating the work table.

Q47. What moves during cutting in a planer?

Answer: The workpiece and table reciprocate during the main cutting motion.

Q48. What is the main use of a planer?

Answer: A planer is used to produce large flat, horizontal, vertical, or inclined surfaces and related profiles.

Q49. Why is a planer used for large workpieces?

Answer: Its rigid construction and large table allow heavy and large components to be machined.

Q50. What is feed in a planer?

Answer: Feed is the transverse or incremental movement of the cutting tool between successive cutting strokes.

Q51. What is a planer table?

Answer: The planer table supports and carries the workpiece during reciprocating motion.

Q52. What is a double housing planer?

Answer: A double housing planer has two vertical housings supporting the cross rail and tool heads, providing rigidity for large work.

Q53. What is the difference between shaper and planer?

Answer: A shaper reciprocates the cutting tool, whereas a planer reciprocates the workpiece and table.

5. Milling

Q54. What is milling?

Answer: Milling is a machining process in which a rotating multi-point cutting tool removes material from a workpiece.

Q55. What is a milling cutter?

Answer: A milling cutter is a rotary multi-tooth cutting tool used to remove material during milling.

Q56. What is face milling?

Answer: Face milling produces a flat surface using cutting edges mainly on the face of the cutter.

Q57. What is slab milling?

Answer: Slab milling produces a flat surface using a cylindrical cutter with teeth on its periphery.

Q58. What is end milling?

Answer: End milling uses an end mill to machine slots, pockets, steps, and profiles.

Q59. What is up milling?

Answer: In up milling, the cutter rotation at the cutting point is opposite to the direction of workpiece feed.

Q60. What is down milling?

Answer: In down milling, the cutter rotation at the cutting point is in the same direction as the workpiece feed.

Q61. What is a vertical milling machine?

Answer: A vertical milling machine has a spindle oriented approximately vertically and is commonly used for face, end, slot, and profile milling.

Q62. What is a horizontal milling machine?

Answer: A horizontal milling machine has a spindle oriented approximately horizontally and is useful for slab milling and other operations.

Q63. What is a workholding device in milling?

Answer: A workholding device securely locates and clamps the workpiece, such as a machine vice or fixture.

Q64. What is a slotting operation?

Answer: Slotting is milling a narrow groove or slot in a workpiece.

Q65. Why is cutting fluid used in milling?

Answer: Cutting fluid can reduce heat and friction, improve tool life, and help remove chips when appropriate.

6. Metal Casting

Q66. What is casting?

Answer: Casting is a manufacturing process in which molten metal is poured into a mould cavity and allowed to solidify.

Q67. What is a mould?

Answer: A mould is a cavity or form that gives the molten metal its required shape during solidification.

Q68. What is a pattern?

Answer: A pattern is a replica or representation used to form the mould cavity into which molten metal is poured.

Q69. What is a sprue?

Answer: A sprue is a vertical passage through which molten metal enters the mould system.

Q70. What is a runner?

Answer: A runner is a passage that carries molten metal from the sprue toward one or more mould cavities.

Q71. What is a gate?

Answer: A gate is the passage through which molten metal enters the mould cavity.

Q72. What is a riser?

Answer: A riser is a reservoir of molten metal that feeds the casting during solidification to compensate for shrinkage.

Q73. What is a casting defect?

Answer: A casting defect is an unwanted imperfection in a cast product, such as porosity, shrinkage, or cold shut.

Q74. What is shrinkage defect?

Answer: Shrinkage occurs when metal contracts during solidification and insufficient molten metal is available to compensate for the contraction.

Q75. What is a core?

Answer: A core is a shaped insert used to produce internal holes, cavities, or passages in a casting.

Q76. Why is draft provided on a pattern?

Answer: Draft provides a slight taper so the pattern can be removed from the mould without damaging the mould cavity.

Q77. What is solidification?

Answer: Solidification is the change of molten metal into a solid state as it loses heat.

7. Sand Moulding

Q78. What is sand moulding?

Answer: Sand moulding is a casting method in which a mould cavity is formed in specially prepared moulding sand.

Q79. What is moulding sand?

Answer: Moulding sand is a prepared granular material used to make a mould cavity for casting.

Q80. What is green sand?

Answer: Green sand is moist moulding sand containing sand, clay, water, and other additives as required.

Q81. What is a flask?

Answer: A flask is a frame or container used to hold and support the moulding sand.

Q82. What is a cope?

Answer: The cope is the upper half of a two-part moulding flask.

Q83. What is a drag?

Answer: The drag is the lower half of a two-part moulding flask.

Q84. What is a parting line?

Answer: The parting line is the boundary where the two halves of the mould meet.

Q85. What is permeability of moulding sand?

Answer: Permeability is the ability of moulding sand to allow gases and air to escape through the mould.

Q86. What is refractoriness?

Answer: Refractoriness is the ability of moulding sand to withstand high temperature without melting or degrading excessively.

Q87. Why is a vent provided in a sand mould?

Answer: A vent allows gases and air to escape from the mould cavity during pouring and solidification.

8. Workshop Safety

Q88. Why is workshop safety important?

Answer: Workshop safety prevents injuries, equipment damage, and unsafe working conditions.

Q89. What PPE is commonly used in a mechanical workshop?

Answer: Common PPE includes safety glasses, suitable footwear, gloves where appropriate, hearing protection, and task-specific protective equipment.

Q90. Why should loose clothing be avoided near rotating machines?

Answer: Loose clothing can become caught in moving parts and create a serious entanglement hazard.

Q91. Why should machine guards not be removed during operation?

Answer: Guards protect users from moving parts, chips, sparks, and other hazards.

Q92. What should be done before maintenance on a machine?

Answer: The machine should be stopped, isolated from its energy sources, and secured according to the applicable safety procedure before maintenance.

Q93. Why should chips not be removed by hand?

Answer: Sharp chips can cut skin; suitable tools such as a brush or chip hook should be used when the machine is stopped as required.

Q94. Why is housekeeping important?

Answer: Good housekeeping reduces slips, trips, fire hazards, and interference around machines.

Q95. Why should a damaged electrical cable not be used?

Answer: A damaged cable can expose users to electric shock, short circuits, or fire hazards.

Q96. What should you do after an accident or near miss?

Answer: Stop the unsafe activity, make the area safe, inform the responsible supervisor or instructor, and follow the institution's reporting procedure.

Q97. What is an emergency stop?

Answer: An emergency stop is a control intended to quickly stop hazardous machine motion when an emergency occurs.

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